Example 4.4. *For the output of full wave rectifier, determine : (i) RMS value (ii) Average value, (iii) Form factor, (iv) Peak factor.*

Solution. The output current of full wave rectifier in shown in Fig. 4.44.

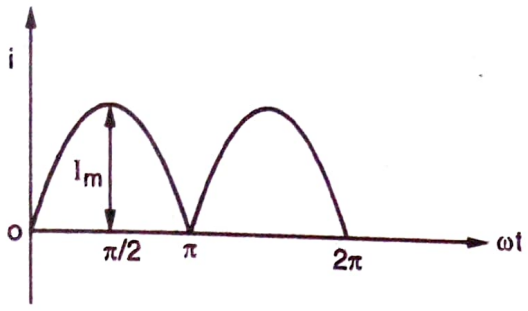


Fig. 4.44.

Time period = π

(*i*) $$I_{rms} = \sqrt{\frac{1}{\pi}\int_0^{\pi} i^2\, d(\omega t)} = \sqrt{\frac{1}{\pi}\int_0^{\pi} I_m^2 \sin^2 \omega t\, d(\omega t)}$$

$$I_{rms} = \sqrt{\frac{I_m^2}{2\pi}\int_0^{\pi} (1 - \cos 2\omega t)\, d(\omega t)} = \frac{I_m}{\sqrt{2}} = 0.707\, I_m.$$

(*ii*) $$I_{av} = \frac{1}{\pi}\int_0^{\pi} I_m \sin \omega t\, d(\omega t) = \frac{I_m}{\pi}[-\cos \omega t]_0^{\pi} = \frac{2\, I_m}{\pi} = 0.637\, I_m$$

(*iii*) **Form factor** $$K_f = \frac{I_{rms}}{I_{av}} = \frac{0.707\, I_m}{0.637\, I_m} = 1.1.$$

(*iv*) **Peak factor** $$K_p = \frac{I_m}{I_{rms}} = \frac{I_m}{0.707\, I_m} = 1.414.$$

Example 4.5. *For the saw tooth wave shown in Fig. 4.45, determine the form factor.*

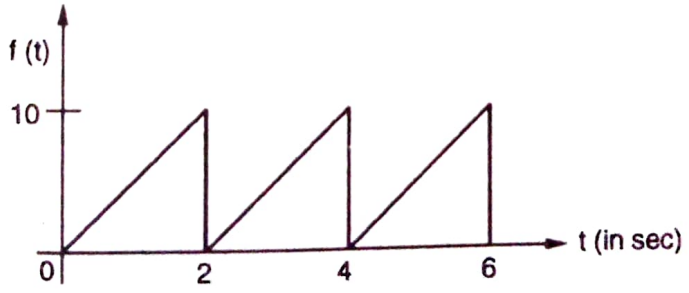


Fig. 4.45.

Solution. Time period of given wave = 2 sec. Expression of function $f(t)$ for first time period

$$f(t) = \frac{10}{2} t = 5t \qquad \text{for } 0 \le t \le 2$$

RMS value :

$$F_{rms} = \sqrt{\frac{1}{T}\int_0^T f^2(t)\,dt} = \sqrt{\frac{1}{2}\int_0^2 (5t)^2\,dt} = \sqrt{\frac{25}{2}\int_0^2 t^2\,dt}$$

$$F_{rms} = \sqrt{\frac{25}{2}\left|\frac{t^3}{3}\right|_0^2} = 5.77.$$

Average value $$F_{av} = \frac{1}{T}\int_0^T f(t)\,dt = \frac{1}{2}\int_0^2 5t\,dt = \frac{5}{2}\left|\frac{t^2}{2}\right|_0^2 = \frac{5}{4}(4) = 5.$$

Form factor $$K_f = \frac{F_{rms}}{F_{av}} = \frac{5.77}{5} = 1.154.$$

Example 4.6. *Find the rms and average value of the wave form shown in Fig. 4.46, where in the first interval* $f(t) = 10e^{-200t}$

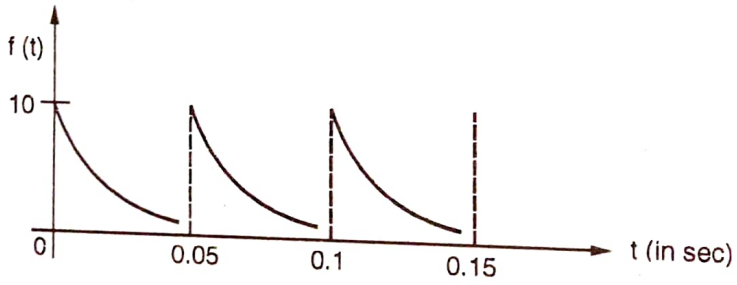


Fig. 4.46.

Solution. RMS value $$F_{rms} = \sqrt{\frac{1}{T}\int_0^T [f(t)]^2\,dt}$$

$$= \sqrt{\frac{1}{0.05}\int_0^{0.05} [10e^{-200t}]^2\,dt} = \sqrt{5} = 2.24$$

Average value $$F_{av} = \frac{1}{T}\int_0^T f(t)\,dt = \frac{1}{0.05}\int_0^{0.05} 10e^{-200t}\,dt$$

$$F_{av} = 1.$$

Example 4.7. *Find the effective (or RMS) value for the voltage function.*

$$v(t) = 20 + 30 \sin \omega t + 60 \sin (2\omega t - \pi/2).$$

Solution. The effective or RMS value of $v(t)$ can be determined by equation (4.13).

$$V_{rms} = \sqrt{(20)^2 + \left(\frac{30}{\sqrt{2}}\right)^2 + \left(\frac{60}{\sqrt{2}}\right)^2}$$

$$= \sqrt{400 + 450 + 1800} = 51.478 \text{ volt.}$$

Example 4.8. *Find rms value, average value, form factor and peak factor for the pulse train shown in Fig. 4.47.*

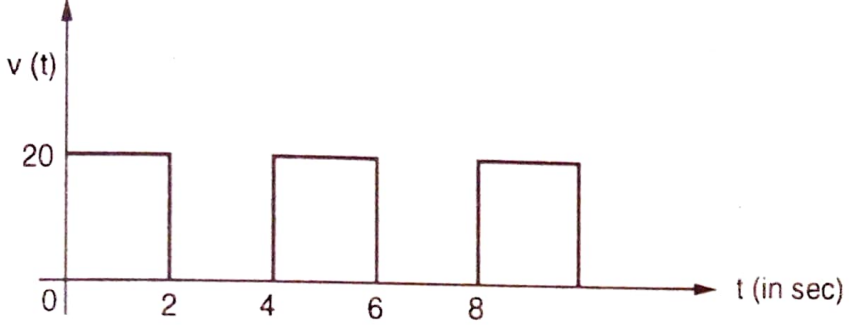


Fig. 4.47.

Solution. Time period of given wave $T = 4$ sec.

RMS value $$V_{rms} = \sqrt{\frac{1}{T}\int_0^T [v(t)]^2\, dt} = \sqrt{\frac{1}{4}\left[\int_0^2 (20)^2\, dt + \int_2^4 0\, dt\right]}$$

$$= \sqrt{\frac{400}{4} | t |_0^2} = 14.142$$

Average value $$V_{av} = \frac{1}{T}\int_0^T v(t)\, dt = \frac{1}{4}\int_0^2 20\, dt$$

$$V_{av} = \frac{20}{4} | t |_0^2 = 10$$

Form factor $$K_f = \frac{V_{rms}}{V_{av}} = \frac{14.142}{10} = 1.414$$

Peak factor. $$K_p = \frac{V_m}{V_{rms}} = \frac{20}{14.142} = 1.414.$$

So the voltage

Example 4.18. *In series R-L circuit, shown in Fig. 4.50 find (i) Impedance, (ii) Resultant current, (iii) Power factor and its nature, (iv) Quality factor.*

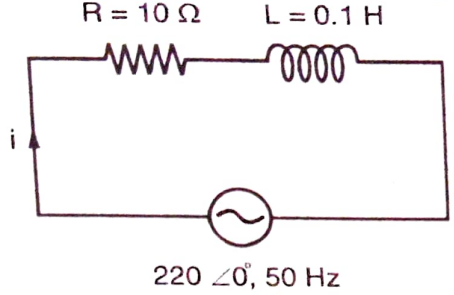


Fig. 4.50.

Solution. The inductive reactance X_L is given by

$$X_L = \omega L = 2\pi f L = 2\pi \times 50 \times 0.1\ \Omega = 31.42\ \Omega.$$

The impedance in rectangular form is given by

$$Z = R + j\,X_L = 10 + j\,31.42\ \ \Omega$$

In polar form $$Z = 32.97 \angle 72.345^\circ\ \ \Omega$$

The resultant current is given by, $$i = \frac{V}{Z} = \frac{220 \angle 0^\circ}{32.97 \angle 72.345^\circ} = 6.672 \angle -72.345^\circ \text{ Amp}$$

The current is lagging since the circuit is inductive.

The power factor angle ϕ, is the angle between voltage and current

$$\phi = 72.345^\circ \text{ (lagging)}$$

$\therefore$ p.f. $\cos \phi = \cos (72.345^\circ) = 0.303$ lag

Alternatively the power factor can be determined by

$$\cos \phi = \frac{R}{Z} = \frac{10}{32.97} = 0.303 \text{ lag}$$

Quality factor is given by

$$Q = \frac{1}{\cos \phi} = 3.3.$$

Example 4.19. *In the given circuit, find the active, reactive and apparant power.*

Solution. The inductive reactance

$$X_L = 2\pi fL = 2\pi \times 60 \times 0.2$$
$$= 75.408\ \Omega$$

Impedance $Z = 15 + j\,75.408\ \Omega$

In polar form $= 76.885 \angle 78.749^\circ\ \Omega$

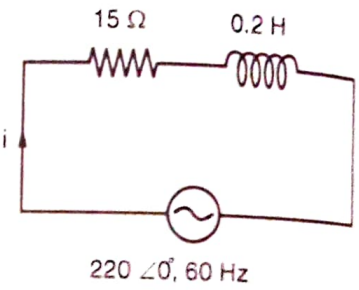


Fig. 4.51.

Current in the circuit :

$$i = \frac{V}{Z} = \frac{220 \angle 0^\circ}{76.885 \angle 78.749^\circ}$$
$$= 2.861 \angle -78.749^\circ \text{ Amp (lagging)}$$

Power factor $\cos \phi = \cos 78.749^\circ = 0.195$ (lag)

and $\sin \phi = \sin 78.749^\circ = 0.98.$

Active power : $P = VI \cos \phi = 220 \times 2.861 \times 0.195 = 122.736$ Watts

or $P = I^2R = 122.77$ Watts.

Reactive power : $Q = VI \sin \phi = 220 \times 2.861 \times 0.98$
$= 616.83$ Volt–Amp reactive

or $Q = I^2X = 617.2$ VAr

Apparant power :

$$S = VI = 220 \times 2.861 = 629.42 \text{ Volt–amp}$$

or $S = \sqrt{P^2 + Q^2} = 629$ Volt–amp.

Example 4.20. *In the capacitive circuit of Fig. 4.52. Find (i) Impedance, (ii) Resultant current, (iii) Power factor, (iv) Power absorbed by the circuit, (v) Phasor relation between different voltages and currents.*

olution. (*i*) The capacitive reactance is given by,

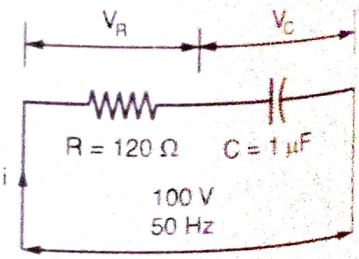


Fig. 4.52

$$X_C = \frac{1}{\omega c} = \frac{1}{2\pi fc}$$

$$= \frac{1}{2\pi \times 50 \times 1 \times 10^{-6}} = 3182.68\ \Omega$$

So the impedance $Z = R - j X_C = 120 - j\,3182.68\ \Omega$

In polar form $Z = 3184.94 \angle -87.84^\circ\ \Omega$

(*ii*) Resultant current

$$i = \frac{V}{Z} = \frac{100\angle 0^\circ}{3184.94 \angle -87.84^\circ} = 0.03139 \angle 87.84^\circ \text{ Amp (leading)}$$

(*iii*) Power factor $\cos \phi = \cos 87.84^\circ = 0.0376$ lead

(*iv*) Power is absorbed by resistor only and that is also known as the active power in the circuit.

$$P = I^2R = (0.03139)^2 \times 120 = 0.1182 \text{ Watts.}$$

(*v*) Voltage drop across resistor :

$$V_R = i\,R = 0.03139 \angle 87.84^\circ \times 120 = 3.7668 \angle 87.84^\circ \text{ volt}$$

(in phase with current i)

Voltage drop across capacitor

$$V_c = i\,(-j X_c) = 0.03139 \angle 87.84^\circ \times 3182.68 \angle -90^\circ$$

$$= 99.9 \angle -2.16^\circ \text{ (lags the } i\text{)}$$

The phasor diagram is shown in Fig. 4.53. Applied voltage of 100 $\angle 0^\circ$ is taken as reference.

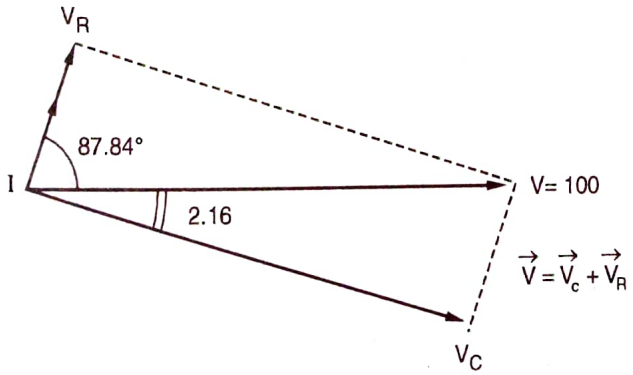


Fig. 4.53.

Example 4.21. *Three voltages represented by* $e_1 = 20 \sin \omega t$, $e_2 = 30 \sin (\omega t - 45)$ *and* $e_3 = \sin(\omega t + 30)$ *are connected in series and then connected to a load of impedance* $(2 + 3j)\ \Omega$. *Find the resultant current and power factor of the circuit. Draw phasor diagram.* *[MACT, Bhopal, 2000 ; RGPV Bhopal, 2001]*

Solution. $e_1 = 20 \sin \omega t$

$e_2 = 30 \sin (\omega t - 45^\circ)$

$e_3 = \sin (\omega t + 30^\circ)$

$$Z = 2 + 3j = \sqrt{2^2 + 3^2} \angle \tan^{-1}\left(\frac{3}{2}\right) = 3.6 \angle 56.3^\circ\ \Omega$$

Resolving the voltage in X-and Y-components

X-component $= 20 \cos 0^\circ + 30 \cos(-45^\circ) + \cos 30^\circ$

$= 20 + 21.2 + 0.866 = 42.06.$

Y-component $= \sin 30^\circ + 30 \sin(-45^\circ) = 0.5 - 21.2 = -20.7.$

Magnitude of resultant voltage

$$V = \sqrt{X^2 + Y^2} = \sqrt{(42.06)^2 + (-20.7)^2} = 46.88\ \text{V}$$

$$\tan \phi = \frac{\text{Y} - \text{component}}{\text{X} - \text{component}} = \frac{-20.7}{42.06} = -0.492$$

$$\phi = -26.2^\circ$$

Resultant voltage $= 46.88 \angle -26.2^\circ$

Wave equation for resultant voltage $= 46.88 \sin(\omega t - 26.2^\circ)$

Circuit current $I = \dfrac{V}{Z} = \dfrac{46.88 \angle -26.2^\circ}{3.6 \angle 56.3^\circ}$

$I = 13.02 \angle -82.5^\circ.$

Power factor of the circuit $= \dfrac{R}{Z} = \dfrac{2}{3.6} = 0.55$ (lagging)

Phasor diagram of the circuit is shown in Fig. 4.54.

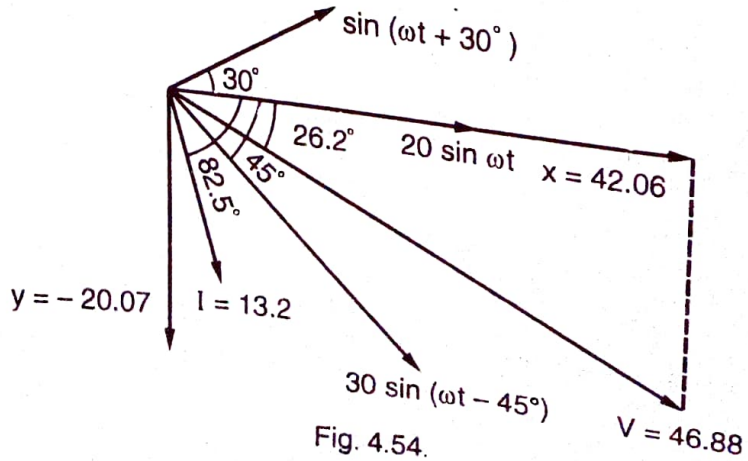


Fig. 4.54.

Example 4.22 *The current in*

Example 4.25. *A 60 Hz sinusoidal voltage $v = 100 \sin \omega t$ is applied to a series R-L circuit. Assuming $R = 10\ \Omega$, $L = 0.01H$, find the steady state current and relative phase angle. Also compute the magnitude and phase of voltage drop across each circuit elements.* *(RGPV. Bhopal, June 2004)*

Solution. The circuit is shown in Fig. 4.56.

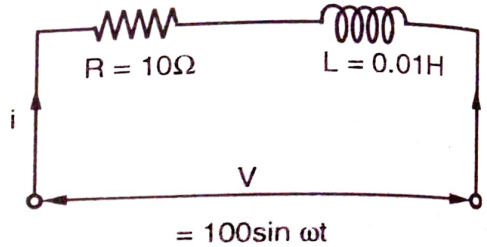


Fig. 4.56.

Given frequency, $f = 60$ Hz

So, $X_L = \omega L = 2\pi fL$

$= 2\pi \times 60 \times 0.01 = 3.77\ \Omega$

Impedance $Z = R + jX_L$

$= 10 + j\,3.77\ \Omega$

In polar form $Z = \sqrt{R^2 + X_L^2} \angle \tan^{-1} X_L/R = 10.687 \angle 20.65^\circ$

$\therefore$ Current $i = \dfrac{v}{z} = \dfrac{100 \sin \omega t}{10.687 \angle 20.65^\circ} = 9.357 \sin (\omega t - 20.65^\circ)$

Current is lagging voltage by angle of 20.65°. We can also write $v = 100 \angle 0^\circ$ volt and $i = 9.357 \angle -20.65^\circ$ Amp.

So. voltage drop across R, $V_R = i\,R = (9.357 \angle -20.65^\circ)(10)$

$= 93.57 \angle -20.65^\circ$ volt

Voltage drop across L, $V_L = j\,i\,X_L = (9.357 \angle -20.65^\circ)(j\,3.77)$

$= (9.357 \angle -20.65^\circ)(3.77 \angle 90^\circ)$

$= 35.275 \angle 69.35^\circ$ volt.

Example 4.26. *A resistor of 15 Ω is connected in series with an inductor of 0.1 H. The circuit is supplied by 230 volt, 50 Hz single phase source. Find the*

(i) Current flowing in the circuit *(ii) Power factor of the circuit*

(iii) Voltage across the reactor *(iv) Active power supplied by source*

(v) Peak current flowing in circuit.

(RGPV, Bhopal, June 2005)

Solution. The circuit is shown in Fig. 4.57.

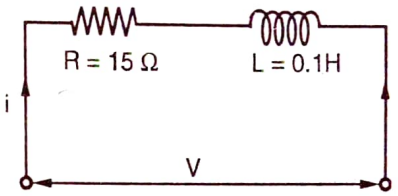


Fig. 4.57.

Given that, $v = 230 \angle 0^\circ$

$f = 50$ Hz

So inductive reactance $X_L = \omega L = 2\pi f L$

$= 2\pi (50)(0.1) = 31.42\ \Omega$

The impedance is given by $Z = R + j\,X_L = 15 + j\,31.42\ \Omega$

Impedance in polar form $Z = 34.816 \angle 64.48^\circ\ \Omega$

(*i*) Current flowing in the circuit

$$i = \frac{V}{Z} = \frac{230 \angle 0^\circ}{34.816 \angle 64.48^\circ} = 6.606 \angle -64.48^\circ \text{ Amp.}$$

(*ii*) Current is lagging by angle 64.48° behind the voltage ($\phi = 64.48^\circ$ lag).

The P.f. of circuit is cos ϕ

$$\text{P.f.} = \cos\phi = \cos 64.48^\circ = 0.43 \text{ lag.}$$

(*iii*) The voltage across reactor or X_L is

$$V_L = i\,(j\,X_L) = (6.606 \angle -64.48^\circ)(j\,31.42)$$

$$= (6.606 \angle -64.48^\circ)(31.42 \angle 90^\circ) = 207.56 \angle 25.52^\circ \text{ Volt}$$

(*iv*) The peak value of current is $I_m = 6.606$ Amp.

Example 4.27. *A resistance of 20 Ω and inductance of 0.2 H and capacitance of 100 μF are connected in series across 220 V, 50 Hz mains. Determine the following :*

(i) Impedance, (ii) Current, (iii) Voltage across R, L and C, (iv) Power in watts, (v) P.f. and the angle of lag.

(RGPV, Bhopal, June 2007)

Solution. The circuit is shown in Fig. 4.58.

Given $v = 220 \angle 0^\circ$ volt, $f = 50$ Hz

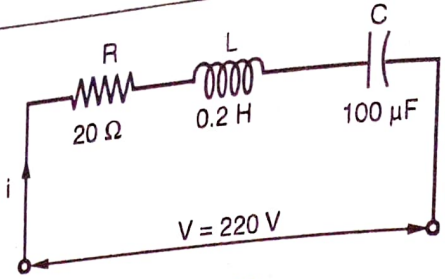


Fig. 4.58.

The inductive reactance,

$$X_L = \omega L = 2\pi f L = 2\pi (50)\ (0.2)$$

$$= 62.84\ \Omega$$

The capacitive reactance, $X_C = \dfrac{1}{\omega C} = \dfrac{1}{2\pi f C} = \dfrac{1}{2\pi (50)\ (100 \times 10^{-6})}$

$$= 31.826\ \Omega.$$

(*i*) Impedance $Z = R + j\,(X_L - X_C)$

$$= 20 + j\,(62.84 - 31.826) = 20 + j\,31.014\ \Omega$$

In polar form $Z = 36.9 \angle 57.183^\circ\ \Omega$

(*ii*) Current $i = \dfrac{V}{Z} = \dfrac{220 \angle 0^\circ}{36.9 \angle 57.183} = 5.962 \angle -57.183^\circ$ Amp

Current is lagging by angle 57.183°.

(*iii*) Voltage across *R*, $V_R = iR = (5.962 \angle 57.183^\circ)\ (20)$

$$= 119.24 \angle -57.183^\circ \quad \text{volt}$$

Voltage across *L*, $V_L = i\,(jX_L) = (5.962 \angle -57.183^\circ)\ (62.84 \angle 90^\circ)$

$$= 374.65 \angle 32.817^\circ \text{ volt}$$

Voltage across *C*, $V_C = i\,(-jX_C) = (5.962 \angle -57.183^\circ)\ (31.826 \angle -90^\circ)$

$$= 189.746 \angle -147.183^\circ \text{ volt.}$$

(*iv*) Power in watts (Active power). This is the power dissipated in resistor of 20 Ω.

$$P = |i|^2 \times 20 = (5.962)^2 \times 20 = 710.9 \text{ watts.}$$

(*v*) **Angle of lag and power factor.** Current is lagging behind the voltage by an angle of 57.183°.

$\therefore$ $\phi = 57.183^\circ$

$$\text{P.f.} = \cos\phi = \cos 57.183^\circ = 0.5419 \text{ lag.}$$

Alternatively the P.f. $= \dfrac{R}{Z} = \dfrac{20}{36.9} = 0.542$ lag.

Example 4.28. *A series R-C circuit takes a power of 7000 W when connected to 200 V, 50 Hz supply. The voltage across the resistor is 130 V. Calculate (i) the resistance (R), current (I), pf (cos ϕ), capacitance (C) and impedance (Z) and (ii) Also write equations for v(t) and i(t).*

(*ii*) Resistance of coil R_L is calculated as,

$$R_L = \sqrt{(Z_L)^2 - (X_L)^2} = \sqrt{(57)^2 - (48.145)^2} = 30.5133\ \Omega.$$ Ans

(*iii*) Resistance of resistor, $R_1 = R - R_L = 63.89 - 30.5133 = 33.38\ \Omega$ **Ans.**

Example 4.32. *An inductive coil of resistance 32 Ω and reactance 15.7 Ω is connected in series with a capacitor of reactance 79.5 Ω The circuit is connected across 500 volt AC supply. Determine (i) Current (ii) Phase difference between applied voltage and current, (iii) voltage across the inductive coil, (iv) Total power absorbed.*

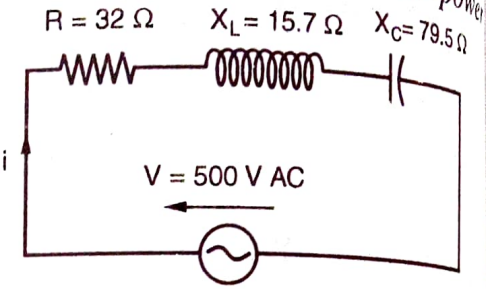


Fig. 4.61.

Solution. $R = 32\ \Omega$,

$$X = X_L - X_C = 15.7 - 79.5 = -63.8\ \Omega$$

(So the circuit is capacitive)

Impedance $Z = R + jX = 32 - 63.8j\ \Omega$

(*i*) Circuit current,

$$i = \frac{V}{Z} = \frac{500\angle 0^\circ}{32 - 63.8j} = 7\angle 63.36^\circ \text{ Amp}$$

(*ii*) Angle between voltage and current, $\phi = 63.36^\circ$ (leading)

(*iii*) Impedance of coil $Z_L = R + jX_L = 32 + j\,15.7\ \Omega$

$$|Z_L| = 35.6439\ \Omega$$

Voltage across coil $= I Z_L = 7 \times 35.6439 = 249.5075$ Volt

Power absorbed $I^2R = (7)^2 \times 32 = 1568$ watts

Example 4.33. *A resistance of 12 Ω, inductance of 0.1 H and a capcitance of 100 μF are connected in series across ac 220 V, 100 Hz supply. Calculate the current and its power factor, power consumed and phase angle mentioning whether it is leading or lagging.*

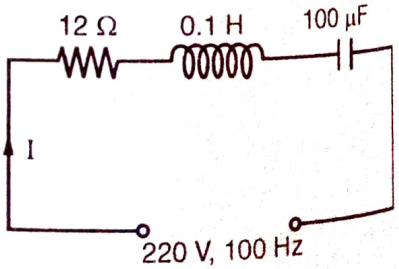


Fig. 4.62.

Solution. Circuit resistance, $R = 12\ \Omega$

Supply frequency, $f = 100$ Hz

Inductive reactance of the circuit, $X_L = 2\pi fL$

$$= 2\pi \times 100 \times 0.1 = 62.83\ \Omega$$

Capacitive reactance of the circuit, $X_C = \frac{1}{2\pi fC}$

Circuit impedance, $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{12^2 + (62.83 - 15.92)^2}$

$= 48.42\ \Omega$ (inductive)

($\because$ inductive reactance exceeds capacitive reactance)

Circuit current, $I = \dfrac{V}{Z} = \dfrac{220}{48.42} = 4.54$ A **Ans.**

Circuit power factor $= \cos\phi = \dfrac{R}{Z} = \dfrac{12}{48.42} = 0.2478$ (lagging) **Ans.**

Power consumed, $P = I^2R = (4.54)^2 \times 12 = 247.7$ Watts **Ans.**

Phase angle, $\phi = \cos^{-1} 0.2478 = 75.65°C$ (lagging) **Ans.** ($\because$ circuit is inductive)

EXAMPLE

Power factor of mains

Example 4.38. *Find i_1, i_2 and i for the circuit shown in Fig. 4.66 and draw the vector diagram.*

(MACT, Bhopal, 1997, M.S. Univ. Baroda)

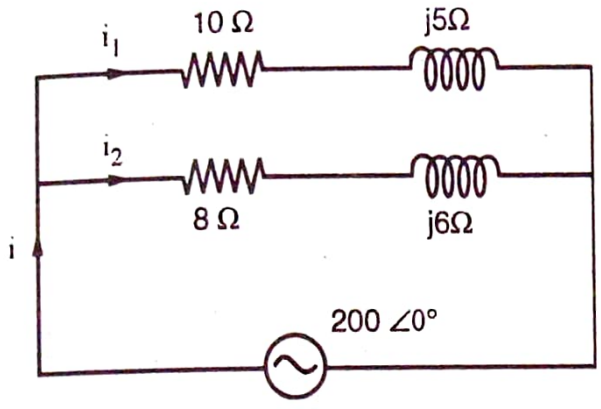


Fig. 4.66.

Solution. Impedance of First branch $= Z_1 = (10 + j\,5)\ \Omega$

Admittance of branch $= Y_1 = \dfrac{1}{Z_1} = \dfrac{1}{10 + j\,5} \times \dfrac{10 - j\,5}{10 - j\,5}$

$= \dfrac{10 - j5}{125}$

$= (0.08 - j\,0.04)$

(*i*) Current through branch $i_1 = VY_1 = 200\,(0.08 - j\,0.04) = (16 - j\,8)\ A$

Impedance of second branch, $Z_2 = (8 + j\,6)\ \Omega$

∴ Admittance of branch $= Y_2 = \dfrac{1}{Z_2}$

$= \dfrac{1}{8 + j\,6} \times \dfrac{8 - j\,6}{8 - j\,6} = \dfrac{8 - j6}{800}$

$= (0.08 - j\,0.06)\ \Omega$

(*ii*) Current through second branch,

$i_2 = VY_2 = 200\,(0.08 - j\,0.06) = (16 - j\,12)\ A$

Total circuit current $i = i_1 + i_2 = 16 - j8 + 16 - j\,12 = (32 - j\,20)\ A$

$= 37.736 \angle 32^\circ$

(*iii*) Vector diagram is shown in Fig. 4.67.

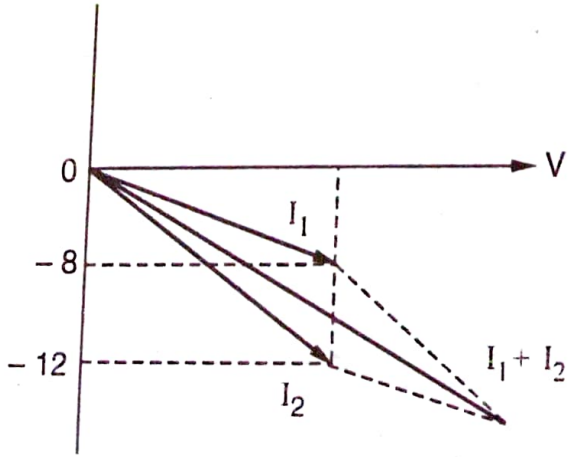


Fig. 4.67. Vector diagram.